ASSIGNMENT-6: SETS

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1. Create an empty set and check its data type and then create a non-empty set with different data items.

In [6]:
```
s1=set()
print(type(s1))
s2={'Srabani', '24BCSB03', 7, 1.1 }
print(s2)
```

```
<class 'set'>
{1.1, '24BCSB03', 'Srabani', 7}
```

2. Create a set and add a single item to it then again add few more items to it.

In [7]:
```
s={2}
s.update(['Srabani', '24BCSB03', 7])
print(s)
```

```
{2, '24BCSB03', 'Srabani', 7}
```

3. Remove the first element of a given set and again from the same set remove any element you want.

In [9]:
```
s={'Srabani', '24BCSB03', 7, 1.1 }
s.remove('Srabani')
print(s)
s.pop()
print(s)
```

```
{1.1, '24BCSB03', 7}
{'24BCSB03', 7}
```

4. Compare between two sets and verify whether a set is a subset of another set or not.

In [13]:
```
s={1,8,6,4,85}
ss={4,6}
if ss.issubset(s):
    print(ss,' is a subset of  ', s)
else:
    print(ss,' is not a subset of  ', s)
```

```
{4, 6}  is a subset of   {1, 4, 85, 6, 8}
```

5. Find the maximum value, minimum value and length of a given set.

In [17]:
```
s={1,8,6,4,8,10,100,5}
print('Length of set: ',len(s))
max_s=max(s)
min_s=min(s)
print(f'Maximum element: {max_s}\nMinimum element: {min_s}')
```

```
Length of set:  7
Maximum element: 100
Minimum element: 1
```

6. Create a set and test whether a given value is present in it or not.

In [20]:
```
s={1,8,6,4,8,10,100,5}
n=int(input('Enter item to test: '))
if n in s:
    print('Present')
else:
    print('Absent')
```

```
Enter item to test: 500
Absent
```

7. Check if two given sets have any common elements in them or not.

In [22]:
```
s1={1,8,6,4,85}
s2={4,6,8,52,4,98,1}
if s1.isdisjoint(s2):
    print('They do not have common elements')
else:
        print('They have common elements')
```

```
They have common elements
```

8. Take two sets and find the numbers missing in second set as compared to the first one and vice versa.

In [23]:
```
s1={1,8,6,4,85}
s2={4,6,8,52,4,98,1}
print('Elements present in s1 but not in s2: ', s1.difference(s2))
print('Elements present in s2 but not in s1: ', s2.difference(s1))
```

```
Elements present in s1 but not in s2:  {85}
Elements present in s2 but not in s1:  {98, 52}
```

9. Find the third largest number from a given set of numbers.

In [31]:
```
s={4,6,8,52,4,98,1}
lar_3=sorted(list(s))[-3]
print('3rd largest: ', lar_3)
```

```
3rd largest:  8
```

10. Test the difference between a set and a frozen set by taking your own example.

In [32]:
```
s = {1,6,3,4,8}
frozen_s = frozenset(s)
s.add(10)
print(s)
frozen_s.add(10)
print(frozen_s)
```

```
{1, 3, 4, 6, 8, 10}
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
~\AppData\Local\Temp\ipykernel_6744\822765942.py in <module>
      3 s.add(10)
      4 print(s)
----> 5 frozen_s.add(10)
      6 print(frozen_s)

AttributeError: 'frozenset' object has no attribute 'add'
```

11. Take a string and check whether it is a binary string or not. [Hint: Binary string - 101010000111]

In [33]:
```
s = input("Enter a string : ")
if (set(s) == {'0', '1'} or set(s) == {'1','0'}):
    print("Binary String.")
else :
    print("Not a binary string.")
```

```
Enter a string : 1010101010
Binary String.
```

12. Use various inbuilt functions of set and do the following task:

a) Return a set of identical items from two given sets.

b) Return a new set will all items from both sets by removing duplicates.

c) Given two sets, update the first set with items that exist only in the first set and not in the second set

d) Remove items from set1 that are not common to both set1 and set2.

e) Update set1 by adding items from set2 except common items.

In [37]:
```
s1 = {1,2,3,4,5}
s2 = {3,4,20}
print(s1.intersection(s2))
```

```
{3, 4}
```

In [36]:
```
print(s1.union(s2))
s1 = s1.difference(s2)
print(s1)
```

```
{1, 2, 3, 4, 5, 20}
{1, 2, 5}
```

In [38]:
```
s1 = {1,2,3,4,5}
s2 = {3,4,20}
s1 = s1.difference(s1.intersection(s2))
print(s1)
```

```
{1, 2, 5}
```

In [39]:
```
s1 = {1,2,3,4,5}
s2 = {4,5,10,12}
s1 = s1.intersection(s2)
print(s1)
```

```
{4, 5}
```

In [40]:
```
s1 = {1,2,3,4,5}
s2 = {4,5,10,12}
```

```python
s1 = s1.union(s2)
print(s1)
```

```
{1, 2, 3, 4, 5, 10, 12}
```